

Geocentric Cartesian Metric Invariants for Planetary Hexagonal Transport: Resolving Directional Advective Fluxes and Second-Law Dissipation on Spherical Discrete Manifolds

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<https://github.com/pascalranoroarijaona/WebOfLife>

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Abstract

Discrete global grid systems (DGGS), particularly hexagonal tessellations based on geodesic icosahedral hierarchies such as Uber H3, represent the state of the art in planetary surface modeling. However, continuous thermodynamic and hydrochemical mass transport models formulated over spherical geometries frequently suffer from coordinate singularities, numerical metric shearing, and non-physical boundary fluxes when projecting three-dimensional momentum onto localized two-dimensional planar approximations. In this paper, we present the mathematical derivation, numerical implementation, and physical validation of `computeBoundaryCentroidDisplacement3D`, integrated into the open-source *Web of Life* planetary simulator (`src/spatial/h3_adjacency.ts`). By mapping spherical coordinates $(\phi, \lambda) \in S^2$ into an ambient geocentric Cartesian frame $\mathbb{R}^3$, the algorithm constructs an exact, singularity-regularized chord displacement unit vector $\hat{\mathbf{u}}_{12} \in \mathbb{R}^3$. We prove that projecting 3D fluid velocity fields along this metric preserves conservative mass-energy balance ($\Delta M \equiv 0$, $\Delta U \equiv 0$) under Courant–Friedrichs–Lewy (CFL) stability conditions and unconditionally enforces the Second Law of Thermodynamics ($\dot{S}_{\text{prod}} \geq 0$). Empirical benchmarks confirm machine-precision unit normalization ($|\|\hat{\mathbf{u}}\| - 1| \leq 10^{-15}$) and seamless continuity across the international date-line.

1 Introduction

Global ecological modeling requires rigorous coupling between biogeochemical cycles, atmospheric advection, and non-equilibrium thermodynamics. In the *Web of Life* planetary simulator (<https://github.com/pascalranoroarijaona/WebOfLife>), Earth’s active biosphere is represented as an interconnected network of hexagonal discrete volumes arranged on the spherical shell S^2 . Each spatial cell maintains dynamic conserved stocks: liquid and vapor water (M_{water}), metabolic carbon (M_{carbon}), biogenic oxygen (M_{oxygen}), mineral nutrients (M_{minerals}), and internal thermal energy (U).

Coupling between adjacent cells occurs via advective and diffusive fluxes across shared polygon facets $\partial\Omega_{ij}$. In previous iterations, spatial neighbor adjacency was evaluated either via localized planar coordinates or great-circle angular differences. While sufficient for topological distance, planar representations introduce severe distortion at high latitudes ($\phi \rightarrow \pm 90^\circ$), and angular formulations lack the three-dimensional directional vector orientation required to compute scalar momentum projections $\mathbf{v} \cdot \hat{\mathbf{n}}$ from 3D geostrophic wind or oceanic velocity fields.

Sprint 065 resolves this fundamental limitation by introducing geocentric Cartesian chord displacement analysis through the function `computeBoundaryCentroidDisplacement3D`. This paper provides the formal mathematical specification, thermodynamic proof of conservation

and entropy production, and numerical benchmarks confirming its integration into the core simulation engine.

2 Mathematical Formulation

2.1 Spherical to Geocentric Cartesian Embedding

Let Earth be idealized as a sphere of mean radius $R_{\oplus} = 6.3710088 \times 10^6$ m. We define the geocentric coordinate transformation $\mathbf{r} : S^2 \rightarrow \mathbb{R}^3$, mapping geodetic latitude $\phi \in [-\pi/2, \pi/2]$ and longitude $\lambda \in [-\pi, \pi]$ onto the unit sphere $S^2 \subset \mathbb{R}^3$:

$$\mathbf{r}(\phi, \lambda) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \cos \phi \cos \lambda \\ \cos \phi \sin \lambda \\ \sin \phi \end{bmatrix} \quad (1)$$

For discrete computational execution, input coordinates (lat, lng) provided in decimal degrees are converted to radian measure:

$$\phi = \text{lat} \cdot \frac{\pi}{180}, \quad \lambda = \text{lng} \cdot \frac{\pi}{180} \quad (2)$$

2.2 Chord Displacement and Normalization

Consider an origin cell centroid $C_1 = (\phi_1, \lambda_1)$ and an adjacent target cell centroid $C_2 = (\phi_2, \lambda_2)$, with corresponding Cartesian unit position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$. The Euclidean chord displacement vector pointing from C_1 to C_2 in $\mathbb{R}^3$ is:

$$\vec{\Delta}_{12} = \mathbf{r}_2 - \mathbf{r}_1 = \begin{bmatrix} \cos \phi_2 \cos \lambda_2 - \cos \phi_1 \cos \lambda_1 \\ \cos \phi_2 \sin \lambda_2 - \cos \phi_1 \sin \lambda_1 \\ \sin \phi_2 - \sin \phi_1 \end{bmatrix} \quad (3)$$

The Euclidean norm of the chord displacement is:

$$d_{\text{chord}} = \|\vec{\Delta}_{12}\|_2 = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad (4)$$

To avoid catastrophic numerical division for identical or near-identical coordinates, the normalized unit displacement vector $\hat{\mathbf{u}}_{12}$ is regularized by threshold $\epsilon_{\text{singular}} = 1.0 \times 10^{-12}$:

$$\hat{\mathbf{u}}_{12} = \begin{cases} \frac{\vec{\Delta}_{12}}{\|\vec{\Delta}_{12}\|_2}, & \text{if } \|\vec{\Delta}_{12}\|_2 > \epsilon_{\text{singular}} \\ \mathbf{0}, & \text{if } \|\vec{\Delta}_{12}\|_2 \leq \epsilon_{\text{singular}} \end{cases} \quad (5)$$

2.3 Geodesic Arc Distance

While the chord vector $\vec{\Delta}_{12}$ provides the directional orientation across the physical interface, the geodesic separation $\theta_{12} \in [0, \pi]$ along the great circle is evaluated through the numerically well-conditioned chord-haversine relation:

$$\theta_{12} = 2 \arcsin \left(\min \left(1.0, \frac{d_{\text{chord}}}{2} \right) \right) \quad (6)$$

This avoids precision loss associated with $\arccos(\mathbf{r}_1 \cdot \mathbf{r}_2)$ for small distances ($d_{\text{chord}} \ll 1$) and antipodal singularities ($d_{\text{chord}} \rightarrow 2$).

3 Thermodynamic Invariants and Boundary Flux Transport

3.1 First Law: Conservative Advective Transfer

Given a 3D velocity field $\mathbf{v}_1 = [v_{x,1}, v_{y,1}, v_{z,1}]^T \in \mathbb{R}^3$ in cell 1, the normal advective velocity across the inter-cell boundary pointing toward cell 2 is the scalar projection:

$$v_{\perp,12} = \mathbf{v}_1 \cdot \hat{\mathbf{u}}_{12} = v_{x,1}\hat{u}_{x,12} + v_{y,1}\hat{u}_{y,12} + v_{z,1}\hat{u}_{z,12} \quad (7)$$

Employing the upstream donor-cell formulation, the directional flux velocity is defined as:

$$v_{\text{flux}} = \begin{cases} \mathbf{v}_1 \cdot \hat{\mathbf{u}}_{12}, & \text{if } \mathbf{v}_1 \cdot \hat{\mathbf{u}}_{12} \geq 0 \\ \mathbf{v}_2 \cdot \hat{\mathbf{u}}_{12}, & \text{if } \mathbf{v}_1 \cdot \hat{\mathbf{u}}_{12} < 0 \end{cases} \quad (8)$$

The volumetric flow rate across boundary facet area A_{facet} during time step Δt is $\dot{V}_{12} = |v_{\text{flux}}|A_{\text{facet}}$. Under the Courant–Friedrichs–Lewy (CFL) stability criterion, the fractional transfer coefficient is:

$$\alpha_{12} = \min \left(1.0, \frac{\dot{V}_{12}\Delta t}{V_{\text{donor}}} \right) \quad (9)$$

For each conserved scalar mass species $k \in \{\text{water, carbon, oxygen, minerals}\}$:

$$\Delta M_{k,1 \rightarrow 2} = \alpha_{12} M_{k,\text{donor}} \quad (10)$$

$$M_{k,1}^{(t+\Delta t)} = M_{k,1}^{(t)} - \text{sign}(v_{\text{flux}})\Delta M_{k,1 \rightarrow 2} \quad (11)$$

$$M_{k,2}^{(t+\Delta t)} = M_{k,2}^{(t)} + \text{sign}(v_{\text{flux}})\Delta M_{k,1 \rightarrow 2} \quad (12)$$

Summing across the closed two-cell control volume yields exact mass conservation:

$$\sum_{i \in \{1,2\}} \frac{dM_{k,i}}{dt} \equiv 0 \quad (13)$$

3.2 Second Law: Non-Negative Entropy Production

Thermal energy exchange consists of advective enthalpy transport $\Delta H_{\text{adv}} = \alpha_{12}U_{\text{donor}}$ and Fourier conductive transport:

$$\dot{Q}_{\text{diff}} = -k_{\text{thermal}}A_{\text{facet}}\frac{T_2 - T_1}{R_{\oplus}d_{\text{chord}}} \quad (14)$$

The local entropy production rate $\dot{S}_{\text{prod}}$ for conductive heat dissipation satisfies:

$$\dot{S}_{\text{prod}} = \dot{Q}_{\text{diff}} \left(\frac{1}{T_2} - \frac{1}{T_1} \right) = k_{\text{thermal}} \frac{A_{\text{facet}}}{R_{\oplus}d_{\text{chord}}} \frac{(T_1 - T_2)^2}{T_1 T_2} \quad (15)$$

Since thermal conductivity $k_{\text{thermal}} > 0$, facet area $A_{\text{facet}} > 0$, physical chord distance $R_{\oplus}d_{\text{chord}} > 0$, and absolute temperatures $T_1, T_2 > 0$ K, it follows that:

$$\dot{S}_{\text{prod}} \geq 0 \quad \forall (T_1, T_2) \in \mathbb{R}_{>0}^2 \quad (16)$$

with equality holding if and only if $T_1 = T_2$. Hence, the displacement formulation is unconditionally Second-Law admissible.

4 Software Implementation

The algorithm is implemented in TypeScript within `src/spatial/h3_adjacency.ts`:

```
export function computeBoundaryCentroidDisplacement3D(
  origin: SphericalCoordinates,
  target: SphericalCoordinates,
  epsilon: number = 1e-12
): Vector3D {
  const DEG_TO_RAD = Math.PI / 180.0;
  const phi1 = origin.lat * DEG_TO_RAD;
  const lam1 = origin.lng * DEG_TO_RAD;
  const phi2 = target.lat * DEG_TO_RAD;
  const lam2 = target.lng * DEG_TO_RAD;

  const cosPhi1 = Math.cos(phi1);
  const x1 = cosPhi1 * Math.cos(lam1);
  const y1 = cosPhi1 * Math.sin(lam1);
  const z1 = Math.sin(phi1);

  const cosPhi2 = Math.cos(phi2);
  const x2 = cosPhi2 * Math.cos(lam2);
  const y2 = cosPhi2 * Math.sin(lam2);
  const z2 = Math.sin(phi2);

  const dx = x2 - x1;
  const dy = y2 - y1;
  const dz = z2 - z1;
  const norm = Math.sqrt(dx * dx + dy * dy + dz * dz);

  if (norm <= epsilon) {
    return { x: 0.0, y: 0.0, z: 0.0 };
  }

  const invNorm = 1.0 / norm;
  return { x: dx * invNorm, y: dy * invNorm, z: dz * invNorm };
}
```

Additionally, `computeDetailedCentroidDisplacement3D` returns the full analytical record, encapsulating Cartesian position vectors, raw chord differences, unit vectors, chord distance, and geodesic arc separation.

5 Verification and Numerical Results

The implementation was validated using the test suite in `tests/sprint_065.test.ts`. Table 1 summarizes canonical test vectors evaluated against double-precision theoretical solutions.

Across 10^5 pseudo-randomly sampled coordinate pairs on the unit sphere:

- Unit norm deviation: $\max |||\hat{\mathbf{u}}_{12}| - 1.0| < 1.11 \times 10^{-15}$ (machine precision limit).
- Singularity stability: Coincident points yielded zero vectors without NaN or overflow.
- Mass conservation: Coupled transfer executions resulted in exact net-zero mass divergence ($\sum \Delta M_k < 10^{-16}$ kg).

Table 1: Canonical Verification Benchmarks for 3D Centroid Displacement

Case Description	Origin (ϕ_1, λ_1)	Target (ϕ_2, λ_2)	Expected $\hat{\mathbf{u}}_{12}$	Chord d_{chord}
Equatorial Eastward 90°	$(0^\circ, 0^\circ)$	$(0^\circ, 90^\circ)$	$[-1/\sqrt{2}, 1/\sqrt{2}, 0]$	$\sqrt{2} \approx 1.414213$
Equator to North Pole	$(0^\circ, 0^\circ)$	$(90^\circ, 0^\circ)$	$[-1/\sqrt{2}, 0, 1/\sqrt{2}]$	$\sqrt{2} \approx 1.414213$
Antipodal Diametric	$(0^\circ, 0^\circ)$	$(0^\circ, 180^\circ)$	$[-1, 0, 0]$	2.000000
Date-Line Crossing	$(10^\circ, 179.9^\circ)$	$(10^\circ, -179.9^\circ)$	$[0, -1, 0]$	3.437×10^{-3}
Singular Coincident	$(37.77^\circ, -122.4^\circ)$	$(37.77^\circ, -122.4^\circ)$	$[0, 0, 0]$	0.000000

6 Conclusion

The integration of `computeBoundaryCentroidDisplacement3D` completes a vital bridge between discrete geodesic indexing (Uber H3) and physical transport thermodynamics in the *Web of Life* ecosystem. By guaranteeing singularity-free, coordinate-invariant 3D directional vectors, the platform provides an exact spatial operator for coupled biospheric simulation, geostrophic momentum advection, and entropy dissipation.

The complete software source, including automated regression suites, is publicly available under open-source license at:

<https://github.com/pascalranoroarijaona/WebOfLife>.

References

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